

Eisenstein-Pythagorean Triples: A Parity Classification on the Hexagonal Lattice

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Abstract

We classify Eisenstein-Pythagorean triples—integer solutions to $a^2 - ab + b^2 = c^2$ —via a geometric construction paralleling the classical Gaussian parametrisation. The main result is a **three-way partition**: non-primitive triples ($\gcd(a, b, c) = 3$) arising from even generators $\gamma = m + n\omega$ with $m + n \equiv 0 \pmod{3}$; canonical primitive triples ($\gcd(a, b, c) = 1$, $a > b$) from odd generators with $m > 2n$; and conjugate primitives, the $a \leftrightarrow b$ images of canonicals. This classification emerges from viewing the parameter space $\mathbb{P}^1(\mathbb{Q})$ as fibred over $\mathbb{Z}[\omega]/\langle\beta\rangle \cong \mathbb{Z}_3$, with the two odd fibres yielding primitive triples and the even fibre yielding non-primitives.

Building on the parity theory of Hadi et al. (2025), we show that triple-primitivity is governed by the even/odd dichotomy, while canonicity arises from the conjugation symmetry $\gamma \leftrightarrow \bar{\gamma}$. The boundary $m = 2n$ (where $a = b$) is necessarily non-primitive, so no self-conjugate primitive triples exist.

The striking parallel with the classical Gaussian theory is shown to be no accident: both are governed by the same pair of arithmetic principles. **Representability**, as in which primes can appear as c , is controlled by *splitting*, determined by quadratic reciprocity. **Primitivity** asking when the triple is coprime, is controlled by *ramification*: reduction modulo the ramified prime $\mathfrak{p}$. The mod-2 condition for Pythagorean triples and the mod-3 condition for Eisenstein triples are thus two instances of the same underlying framework.

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1 Introduction

The Pythagorean equation $x^2 + y^2 = z^2$ has been understood since antiquity, but its deeper structure emerges through the arithmetic of the Gaussian integers $\mathbb{Z}[i]$. Primitive triples arise as squares of primitive Gaussian integers, with a binary parity condition ($p \not\equiv q \pmod{2}$) ensuring coprimality.

The square lattice of $\mathbb{Z}[i]$ is not unique, however. The plane admits exactly two regular lattices supporting a Euclidean domain: the square lattice (Gaussian) and the hexagonal lattice (Eisenstein). Moving to the Eisenstein integers $\mathbb{Z}[\omega]$, where $\omega = e^{2\pi i/3}$, the quadratic form becomes $a^2 - ab + b^2$, and the Pythagorean equation transforms into:

$$a^2 - ab + b^2 = c^2.$$

The key algebraic shift is from the “even prime” $1 + i$ (norm 2) to $\beta = 1 - \omega$ (norm 3). This changes the parity classification from binary to ternary, with three cosets in $\mathbb{Z}[\omega]/\langle\beta\rangle \cong \mathbb{Z}_3$. The richer structure leads to a **three-way partition** of Eisenstein-Pythagorean triples:

- (i) **Non-primitive triples** (ntrim): $\gcd(a, b, c) = 3$, from even generators.
- (ii) **Canonical primitive triples** (ctrim): $\gcd(a, b, c) = 1$ with $a > b$, from odd generators.
- (iii) **Conjugate primitive triples**: the $a \leftrightarrow b$ images of canonicals.

Table 1: Two Pythagorean theories: parallel or unified?

	Gaussian $\mathbb{Z}[i]$	Eisenstein $\mathbb{Z}[\omega]$
Lattice	Square	Hexagonal
Norm form	$a^2 + b^2$	$a^2 - ab + b^2$
“Even prime”	$1 + i$, norm 2	$1 - \omega$, norm 3
Parity classes	2	3
Primitivity test	$p \not\equiv q \pmod{2}$	$m + n \not\equiv 0 \pmod{3}$

The structural parallels are striking. But *why* should the mod-2 condition in the Gaussian case mirror the mod-3 condition in the Eisenstein case? Is this mere analogy, or is there a deeper principle at work? We will show that both are instances of a single arithmetic phenomenon: **the ramified prime controls primitivity**. The apparent parallel is, in fact, a theorem.

1.1 Outline

Section 2 reviews the hexagonal lattice and Eisenstein integers. Section 3 introduces the unit curve $X^2 - XY + Y^2 = 1$ and its matrix representation. Section 4 develops the six-fold symmetry of the unit group. Section 5 presents the main classification theorem. Section 6–7 states the governing parity conditions and identifies the canonical triples while the projective fibration and coset correspondence are developed in Section 8. The complete sieve and classification tables are in Section 9 and we conclude in Section 10. An interactive companion tool for exploring the classification and the unit elliptical construction is available at [2].

1.2 Rational Points on the Unit Circle

The complete parametrisation of the Pythagorean equation $x^2 + y^2 = z^2$ primitive solutions emerges elegantly from two perspectives: the geometry of the unit circle and the arithmetic of the Gaussian integers. Finding integer solutions to $x^2 + y^2 = z^2$ is equivalent to finding rational points on the unit circle $X^2 + Y^2 = 1$, via $(x, y, z) \leftrightarrow (x/z, y/z)$. This is the standard rational parametrization of the circle obtained by intersecting with the pencil of lines through a rational point.[8]

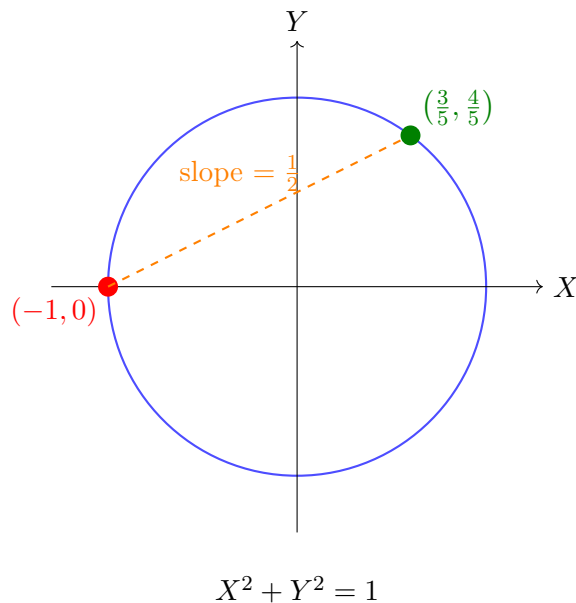


Figure 1: Lines through $(-1, 0)$ with rational slope $t = q/p$ intersect the unit circle at rational points, yielding Pythagorean triples.

Theorem (Classical Parametrisation). *The line $Y = t(X + 1)$ through $(-1, 0)$ with slope $t \in \mathbb{Q}$ intersects the unit circle at*

$$(X, Y) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right).$$

Setting $t = q/p$ with $\gcd(p, q) = 1$ and clearing denominators yields the Pythagorean triple

$$(a, b, c) = (|p^2 - q^2|, 2pq, p^2 + q^2)$$

for $p > q > 0$ with $p \not\equiv q \pmod{2}$.

1.3 The Gaussian Integer Perspective

The parametrisation acquires deeper meaning in $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$.

The Gaussian Squaring Principle

If $\gamma = p + qi$ is a primitive Gaussian integer (i.e., $\gcd(p, q) = 1$), then:

$$\gamma^2 = (p^2 - q^2) + 2pq \cdot i, \quad N(\gamma^2) = N(\gamma)^2 = (p^2 + q^2)^2.$$

The triple $(|p^2 - q^2|, 2pq, p^2 + q^2)$ satisfies $a^2 + b^2 = c^2$.

The moral: Pythagorean triples are squares of primitive Gaussian integers.

The parity condition $p \not\equiv q \pmod{2}$ ensures the triple is primitive: when $p \equiv q \pmod{2}$, both are odd, and $2 \mid \gcd(a, b, c)$. **Our goal:** Develop the complete analogue for the Eisenstein integers $\mathbb{Z}[\omega]$, where the “even prime” 2 is replaced by $\beta = 1 - \omega$ with $N_\beta(\beta) = 3$. This leads to a mod-3 parity classification and a richer three-way partition of triples.

2 The Hexagonal Lattice and Eisenstein Integers

The Gaussian integers live on a square lattice, while the Eisenstein integers live on a hexagonal lattice. This geometric shift changes the relevant quadratic form from $x^2 + y^2$ to $a^2 - ab + b^2$. Take thus, the unit basis vectors $\mathbf{e}_1, \mathbf{e}_2$ meeting at 60° . A point (x, y) represents the vector $\mathbf{v} = x\mathbf{e}_1 + y\mathbf{e}_2$, with squared length given by the parallelogram law:

$$|\mathbf{v}|^2 = x^2 + y^2 + 2xy \cos(60^\circ) = x^2 + xy + y^2.$$

The squared distance from the origin to a lattice point (x, y) is given by the **Loeschian form**:

$$L(x, y) = x^2 + xy + y^2.$$

A positive integer representable by this form is called a **Loeschian number**. The points where $L(x, y)$ is a perfect square. That is, potential Eisenstein-Pythagorean solutions which lie along three distinguished directions.

Example. $L(3, 2) = 9 + 6 + 4 = 19$ (Loeschian, since $19 \equiv 1 \pmod{3}$). But 17 and 18 are not Loeschian, being “forbidden distances” on the hexagonal lattice as no point lies at these distances from the origin. The next achievable distance after $\sqrt{16} = 4$ is $\sqrt{19}$.

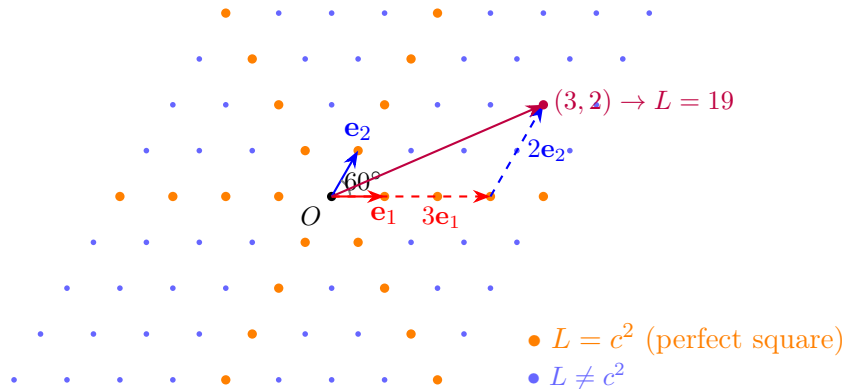


Figure 2: The hexagonal lattice with basis vectors $\mathbf{e}_1, \mathbf{e}_2$ at 60° . The point $(3, 2)$ means $3\mathbf{e}_1 + 2\mathbf{e}_2$, giving $L(3, 2) = 9 + 6 + 4 = 19$. Orange points lie along the lattice axes where $L = c^2$.

Proposition (Loeschian Characterisation). *A positive integer n is Loeschian iff every prime $p \equiv 2 \pmod{3}$ dividing n appears to an even power.*

2.1 The Eisenstein Integers

The same lattice admits a second coordinate system using $\omega = e^{2\pi i/3} = \frac{-1+i\sqrt{3}}{2}$, with basis angle 120° instead of 60° .

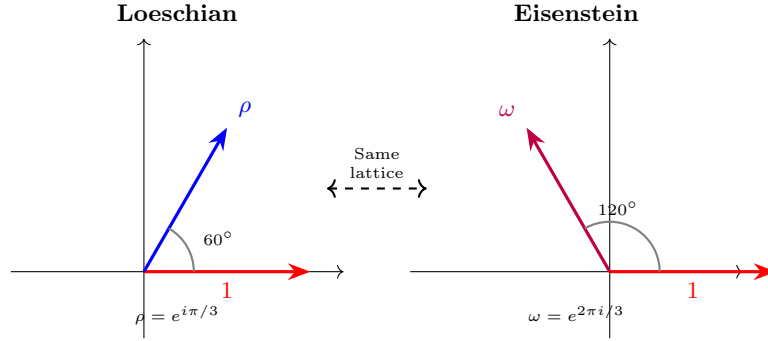


Figure 3: Two coordinate frames for the hexagonal lattice.

The **Eisenstein integers** are $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$. The **Eisenstein norm** is

$$N_\rho(a + b\omega) = (a + b\omega)(a + b\bar{\omega}) = a^2 - ab + b^2.$$

The sign change from $+xy$ to $-ab$ reflects angle change: $\cos(60^\circ) = +\frac{1}{2}$ versus $\cos(120^\circ) = -\frac{1}{2}$. The following table summarises the key structural differences between the two Euclidean domains.

Table 2: One lattice, two coordinate systems

	Loeschian	Eisenstein
Basis angle	60°	120°
Form	$x^2 + xy + y^2$	$a^2 - ab + b^2$
Transformation	$(x, y) \leftrightarrow (a, -b)$	
Discriminant	-3	

Table 3: Comparison of Gaussian and Eisenstein integers

Property	Gaussian $\mathbb{Z}[i]$	Eisenstein $\mathbb{Z}[\omega]$
Lattice	Square	Hexagonal
Norm	$p^2 + q^2$	$m^2 - mn + n^2$
Units	$\{\pm 1, \pm i\}$ (4)	$\{\pm 1, \pm \omega, \pm \omega^2\}$ (6)
“Even prime”	$1 + i$, norm 2	$\beta = 1 - \omega$, norm 3
Parity classes	2 (even/odd)	3 (even/odd-1/odd-2)

The algebraic results overleaf from Hadi et al provide the structural backbone for the parity sieve and explain why the generator parity completely determines triple primitivity.

Algebraic foundation: primitive Eisenstein integers. We summarise the algebraic theory from Hadi et al. [1] that underpins the geometric classification developed above.

Definition (Three levels of primitivity). We distinguish three notions:

1. **Generator primitive (prim):** $\gamma = m + n\omega$ with $\gcd(m, n) = 1$.
2. **Output primitive:** $\alpha = \gamma^2 = a + b\omega$ with $\gcd(a, b) = 1$.
3. **Triple primitive (tprim):** (a, b, c) with $\gcd(a, b, c) = 1$.

Remark (Implications). $\text{prim} \implies \text{output primitive} \not\Rightarrow \text{tprim}$ The gap is filled by parity: $\text{output primitive} + \text{odd generator} \implies \text{tprim}$.

Primitive versus prime & quotient isomorphism

Remark (Hadi et al.)

“Primitive” means coordinate-coprime, **not** prime. For example:

- $5 + 4\omega$ is primitive ($\gcd(5, 4) = 1$) but not prime: $5 + 4\omega = (1 - \omega)(2 + 3\omega)$.
- 5 is Eisenstein prime ($5 \equiv 2 \pmod{3}$, inert) but not primitive: $\gcd(5, 0) = 5$.

Theorem 8 (Hadi et al.)

If η is a primitive Eisenstein integer, then: $\mathbb{Z}[\omega]/\langle\eta\rangle \cong \mathbb{Z}_{N_\rho(\eta)}$.

This clean quotient structure explains why primitivity is central in coding theory and why the parity fibration over $\mathbb{Z}_3$ respects algebraic structure.

Characterisation of primitive elements.

Theorem 6 (Hadi et al.)

An Eisenstein integer η is primitive iff for $r \in \{0, 1\}$: $\eta \sim \beta^r \psi_1^{r_1} \cdots \psi_k^{r_k}$, each ψ_i is prime with $N_\rho(\psi_i) = q_i \equiv 1 \pmod{3}$, and the q_i are distinct.

Example (Classifying primitivity).

η	Factorisation	Primitive?	Reason
$3 + \omega$	prime, $N = 7$	Yes	Single split prime
$8 + 5\omega$	$(3 + \omega)^2$	Yes	Repeated split prime OK
$5 + 4\omega$	$(1 - \omega)(2 + 3\omega)$	Yes	$\beta \cdot \psi$
5	inert prime	No	$\gcd(5, 0) = 5$
$6 + 3\omega$	$3(2 + \omega)$	No	Rational factor 3
$(1 - \omega)^2$	$\beta^2 = -3\omega$	No	$\gcd(0, 3) = 3$

Primitivity of squares.

Theorem 5 (Hadi et al.)

If ψ is Eisenstein prime with $N_\rho(\psi) = q \equiv 1 \pmod{3}$, ψ^n is primitive $\forall n \geq 1$.

Corollary. If $\gamma = m + n\omega$ is primitive with $\gcd(m, n) = 1$ and γ contains no inert prime factors, then γ^2 is also primitive.

3 The Eisenstein Unit Curve

Before developing the classification machinery, we establish the geometric arena: the curve defined by the Eisenstein norm. That is, the Eisenstein analogue of the unit circle is the **norm-one curve**:

$$\mathcal{E} : X^2 - XY + Y^2 = 1. \quad (1)$$

This is the locus of points (X, Y) with Eisenstein norm equal to 1.

Proposition. *The curve $\mathcal{E}$ is an ellipse. In matrix form:*

$$(X \ Y) \underbrace{\begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}}_E \begin{pmatrix} X \\ Y \end{pmatrix} = 1.$$

The matrix E has eigenvalues $\lambda_1 = \frac{1}{2}$, $\lambda_2 = \frac{3}{2}$, with eigenvectors along directions 45° and 135° .

Proof. The characteristic polynomial is $\det(E - \lambda I) = (1 - \lambda)^2 - \frac{1}{4} = \lambda^2 - 2\lambda + \frac{3}{4}$, with roots $\lambda = 1 \pm \frac{1}{2}$. $\square$

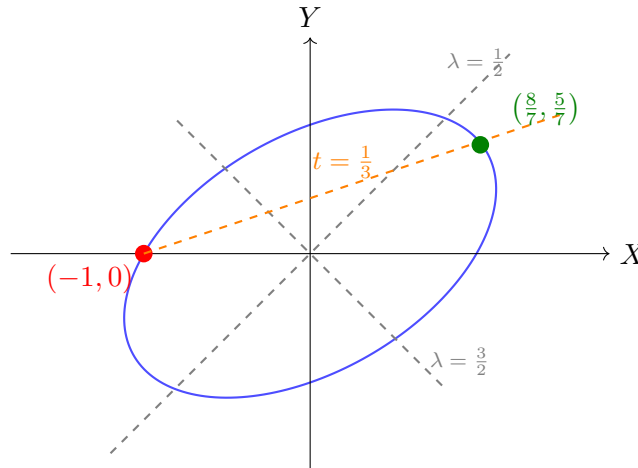


Figure 4: The Eisenstein unit curve $X^2 - XY + Y^2 = 1$ with principal axes shown. Lines through $(-1, 0)$ with rational slope yield Eisenstein-Pythagorean triples.

3.1 The Quadratic Form and Its Transformations

The matrix E encodes the Eisenstein norm. The relationship to the Loeschian form is a simple coordinate transformation.

Proposition (Coordinate Duality). *Let $L = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}$ be the Loeschian matrix (for $x^2 + xy + y^2$) and $E = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$ the Eisenstein matrix. Then:*

$$E = R L R^T, \quad \text{where } R = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The matrix R is the reflection $(x, y) \mapsto (x, -y)$, corresponding to the coordinate change between the 60° and 120° frames.

Proof. Direct computation:

$$RLR^T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix} = E. \quad \square$$

Remark. Both L and E have the same eigenvalues $\{\frac{1}{2}, \frac{3}{2}\}$ and the same determinant $\det(L) = \det(E) = \frac{3}{4}$. The discriminant $\Delta = -3$ (computed as $B^2 - 4AC$ for $Ax^2 + Bxy + Cy^2$) is preserved. The reflection R is an isometry of the underlying lattice.

3.2 Rational Points and the Parametrisation

Theorem (Rational Points on $\mathcal{E}$). *The line $Y = t(X + 1)$ through $(-1, 0)$ with slope $t \in \mathbb{Q}$ intersects $\mathcal{E}$ at:*

$$(X, Y) = \left(\frac{1 - t^2}{1 - t + t^2}, \frac{t(2 - t)}{1 - t + t^2} \right).$$

Setting $t = n/m$ with $\gcd(m, n) = 1$ and clearing denominators yields the Eisenstein-Pythagorean triple:

$$(a, b, c) = (m^2 - n^2, n(2m - n), m^2 - mn + n^2).$$

Proof. Substituting $Y = t(X + 1)$ into $X^2 - XY + Y^2 = 1$:

$$\begin{aligned} X^2 - X \cdot t(X + 1) + t^2(X + 1)^2 &= 1 \\ X^2(1 - t + t^2) + X(2t^2 - t) + (t^2 - 1) &= 0. \end{aligned}$$

One root is $X = -1$ (the base point). By Vieta's formulas, the product of roots is $(t^2 - 1)/(1 - t + t^2)$, so the other root is $X = (1 - t^2)/(1 - t + t^2)$. Then $Y = t(X + 1)$ gives the stated formula. $\square$

This establishes the ellipse as the geometric source of all Eisenstein-Pythagorean triples, paralleling the role of the unit circle in the classical theory.

4 The Six-Fold Symmetry

The hexagonal lattice has 60° rotational symmetry, reflected in the six-element unit group of $\mathbb{Z}[\omega]$. This symmetry acts on the curve $\mathcal{E}$ and on Eisenstein-Pythagorean triples, reducing the effective parameter space. The units of $\mathbb{Z}[\omega]$ are $\mathbb{Z}[\omega]^\times = \{\pm 1, \pm\omega, \pm\omega^2\}$ corresponding to rotations of the hexagonal lattice.

Proposition. *Multiplication by ω acts on $\mathbb{Z}[\omega]$ as a 60° rotation. In matrix form, with respect to the basis $\{1, \omega\}$:*

$$\omega \cdot (a + b\omega) = -b + (a - b)\omega,$$

corresponding to the matrix:

$$M_\omega = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

This matrix satisfies $M_\omega^6 = I$ and $M_\omega^3 = -I$.

Proof. Using $\omega^2 = -1 - \omega$:

$$\omega(a + b\omega) = a\omega + b\omega^2 = a\omega + b(-1 - \omega) = -b + (a - b)\omega.$$

Thus $\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} -b \\ a-b \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}$. $\square$

The six units give rise to the rotation matrices:

Unit u	Matrix M_u	Rotation angle
1	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	0°
ω	$\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$	60°
ω^2	$\begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}$	120°
-1	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	180°
$-\omega$	$\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$	240°
$-\omega^2$	$\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$	300°

The unit group acts on the curve $\mathcal{E}$ by matrix multiplication, preserving the norm.

Proposition. *If (X, Y) lies on $\mathcal{E}$, then so does $M_u \begin{pmatrix} X \\ Y \end{pmatrix}$ for each unit u .*

Proof. The Eisenstein norm is $N_\rho(a + b\omega) = |a + b\omega|^2$. Since $|u| = 1$ for units, $N_\rho(u \cdot \alpha) = |u|^2 N_\rho(\alpha) = N_\rho(\alpha)$. $\square$

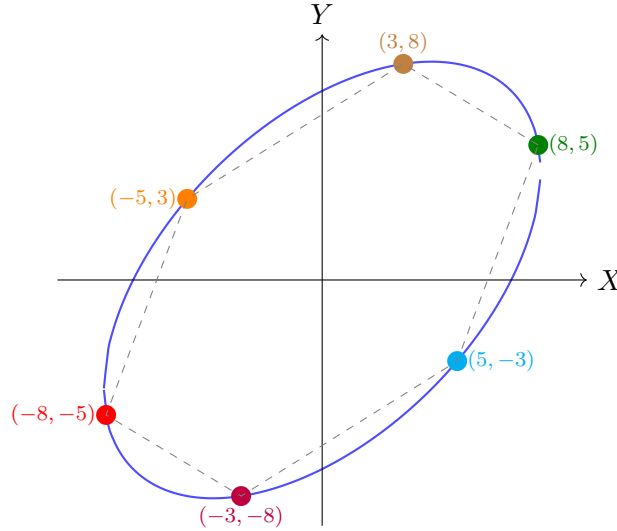


Figure 5: The six images of $(8, 5)/7$ under the unit group action on the Eisenstein unit ellipse $X^2 - XY + Y^2 = 1$. All six points lie on the ellipse and correspond to the same value $c = 7$.

The unit orbit is a regular hexagon in the Eisenstein (Loeschian) metric; when plotted in (a, b) -coordinates it appears skewed because the Euclidean metric in the (a, b) -plane is not the norm metric.

Proposition. *The unit u sends the triple (a, b, c) arising from $\alpha = a + b\omega$ to:*

Unit u	$u \cdot \alpha$	Triple
1	$a + b\omega$	(a, b, c)
ω	$-b + (a - b)\omega$	$(-b, a - b, c)$
ω^2	$(b - a) - a\omega$	$(b - a, -a, c)$
-1	$-a - b\omega$	$(-a, -b, c)$
$-\omega$	$b + (b - a)\omega$	$(b, b - a, c)$
$-\omega^2$	$(a - b) + a\omega$	$(a - b, a, c)$

All six triples satisfy $a^2 - ab + b^2 = c^2$ with the same c .

Example. For $(a, b, c) = (8, 5, 7)$, the orbit consists of:

$$(8, 5, 7), (-5, 3, 7), (-3, -8, 7), (-8, -5, 7), (5, -3, 7), (3, 8, 7).$$

The positive representatives with $a, b > 0$ are $(8, 5, 7)$, $(5, 3, 7)$, and $(3, 8, 7)$.

The six-fold symmetry means each projective point $[a : b : c] \in \mathcal{E}(\mathbb{Q})$ has six affine representatives (up to sign). To obtain unique representatives, we restrict the parameter space.

Definition. The **fundamental domain** for the parametrisation is:

$$\mathcal{F} = \{(m, n) \in \mathbb{Z}^2 : \gcd(m, n) = 1, m > n > 0\}.$$

This selects one representative from each unit group orbit.

Remark. The constraint $m > n > 0$ is equivalent to choosing the “first sextant” of the parameter space. Alternative choices (e.g., $a > b > 0$) would select a different fundamental domain but yield the same set of primitive triples.

5 The Main Classification Theorem

We now state the paper’s central result, then develop the theory to prove it.

5.1 Eisenstein-Pythagorean Triples

Definition. An **Eisenstein-Pythagorean triple** (E-P triple) is a triple (a, b, c) of positive integers satisfying

$$a^2 - ab + b^2 = c^2.$$

The triple is **primitive** (or **triple-primitive**, abbreviated **tprim**) if $\gcd(a, b, c) = 1$.

Theorem (Eisenstein Squaring Principle). *Let $\gamma = m + n\omega$ with $\gcd(m, n) = 1$ and $m > n > 0$. Then γ^2 yields the E-P triple:*

$$(a, b, c) = (m^2 - n^2, n(2m - n), m^2 - mn + n^2) \quad (2)$$

Proof. Using $\omega^2 = -1 - \omega$:

$$(m + n\omega)^2 = m^2 + 2mn\omega + n^2\omega^2 = (m^2 - n^2) + n(2m - n)\omega.$$

Then $c = N_\rho(\gamma) = m^2 - mn + n^2$, and $N_\rho(\gamma^2) = c^2$, giving $a^2 - ab + b^2 = c^2$. $\square$

Main Classification Theorem

The parametrisation (2) on $\{(m, n) : \gcd(m, n) = 1, m > n > 0\}$ induces a **three-way partition**, each class containing asymptotically $\frac{1}{3}$ of parameter pairs:

Class	Condition	Properties
Non-primitive (ntrim)	$m + n \equiv 0 \pmod{3}$	$\gcd(a, b, c) = 3$
Canonical primitive (ctrim)	$m + n \not\equiv 0, m > 2n$	$\gcd(a, b, c) = 1, a > b$
Conjugate primitive	$m + n \not\equiv 0, m < 2n$	$\gcd(a, b, c) = 1, a < b$

Boundary $m = 2n$ forces $m + n = 3n \equiv 0 \pmod{3}$, hence falls into non-primitive class.

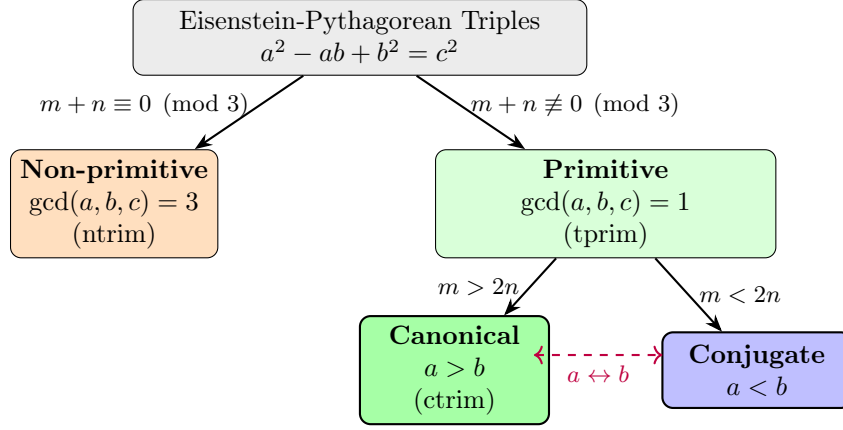


Figure 6: The lineage of Eisenstein-Pythagorean triples. Primitivity is governed by parity; canonicity by the ratio $m : 2n$.

5.2 Discovery: The Pattern in Data

Before proving the theorem, let us observe the pattern empirically.

Table 4: Eisenstein-Pythagorean triples from small parameters

m	n	a	b	c	$m + n$	$m \stackrel{?}{>} 2n$	$\gcd(a, b, c)$
2	1	3	3	3	$3 \equiv 0$	$=$	3
3	1	8	5	7	$4 \equiv 1$	$>$	1
3	2	5	8	7	$5 \equiv 2$	$<$	1
4	1	15	7	13	$5 \equiv 2$	$>$	1
4	3	7	15	13	$7 \equiv 1$	$<$	1
5	1	24	9	21	$6 \equiv 0$	$>$	3
5	2	21	16	19	$7 \equiv 1$	$>$	1
5	3	16	21	19	$8 \equiv 2$	$<$	1
5	4	9	24	21	$9 \equiv 0$	$<$	3
6	1	35	11	31	$7 \equiv 1$	$>$	1
6	5	11	35	31	$11 \equiv 2$	$<$	1
7	2	45	24	39	$9 \equiv 0$	$>$	3

Observations:

1. $\gcd(a, b, c) = 1$ precisely when $m + n \not\equiv 0 \pmod{3}$.
2. $a > b$ precisely when $m > 2n$.
3. Pairs like $(3, 1)$ and $(3, 2)$ produce $(8, 5, 7)$ and $(5, 8, 7)$ —the same triple with $a \leftrightarrow b$.

6 The Parity Classification

The mod-3 pattern reflects a fundamental trichotomy in $\mathbb{Z}[\omega]$, developed by Hadi et al. [1].

6.1 Even and Odd Eisenstein Integers

Definition (Hadi et al., Section 2.4)

Let $\beta = 1 - \omega$. An Eisenstein integer α is:

- **Even** if $\beta \mid \alpha$
- **Odd** if $\beta \nmid \alpha$

Since $N_\rho(\beta) = 3$, the quotient $\mathbb{Z}[\omega]/\langle\beta\rangle$ has exactly three residue classes, represented by $0, 1, 2$, hence $\mathbb{Z}[\omega]/\langle\beta\rangle \cong \mathbb{Z}/3\mathbb{Z}$. [9]

Theorem 10 (Hadi et al.)

For $\alpha = a + b\omega \in \mathbb{Z}[\omega]$:

1. α is **even** $\iff a + b \equiv 0 \pmod{3}$
2. α is **odd Type-1** $\iff a + b \equiv 1 \pmod{3}$
3. α is **odd Type-2** $\iff a + b \equiv 2 \pmod{3}$

Moreover, $N_\rho(\alpha) \equiv 0 \pmod{3}$ iff α is even.

6.2 Why Parity Governs Triple-Primitivity

The key connection is coprimality with conjugates.

Proposition 1 (Hadi et al.)

For $\alpha = a + b\omega$ with $\gcd(a, b) = 1$:

$$\gcd(\alpha, \bar{\alpha}) = \begin{cases} 1 & \text{if } \alpha \text{ is odd} \\ \beta & \text{if } \alpha \text{ is even} \end{cases}$$

Theorem (The Parity Sieve). *Let $\gamma = m + n\omega$ with $\gcd(m, n) = 1$. The triple (a, b, c) from (2) satisfies:*

$$\gcd(a, b, c) = \begin{cases} 1 & \text{if } m + n \not\equiv 0 \pmod{3} \quad (\gamma \text{ odd}) \\ 3 & \text{if } m + n \equiv 0 \pmod{3} \quad (\gamma \text{ even}) \end{cases}$$

Proof. When γ is even, $\beta \mid \gamma$ and $\beta \mid \bar{\gamma}$ (since $\bar{\beta} \sim \beta$). Thus:

- $\beta^2 \mid \gamma^2 = a + b\omega$, so $3 \mid a$ and $3 \mid b$ (since $\beta^2 = -3\omega$).
- $\beta^2 \mid \gamma\bar{\gamma} = c$, so $3 \mid c$.

Hence $3 \mid \gcd(a, b, c)$.

When γ is odd, $\gcd(\gamma, \bar{\gamma}) = 1$ by Proposition 1. Any prime $p \mid \gcd(a, b, c)$ would divide both γ^2 and $\gamma\bar{\gamma}$, hence would divide $\gcd(\gamma, \bar{\gamma})^2 = 1$, a contradiction. $\square$

6.3 Parity Multiplication and the Squaring Map

Theorem 11 (Hadi et al.)

The parity classes multiply as:

$\times$	Even	Odd-1	Odd-2
Even	Even	Even	Even
Odd-1	Even	Odd-1	Odd-2
Odd-2	Even	Odd-2	Odd-1

Corollary (Parity Collapse Under Squaring). *For any $\gamma \in \mathbb{Z}[\omega]$:*

$$\gamma^2 \text{ is } \begin{cases} \text{Even} & \text{if } \gamma \text{ is even} \\ \text{Odd-1} & \text{if } \gamma \text{ is odd (either type)} \end{cases}$$

This explains why all primitive triples have $a + b \equiv 1 \pmod{3}$: both Odd-1 and Odd-2 generators square to Odd-1.

7 Canonical/Conjugate Distinction

Among primitive triples, we distinguish **canonical** representatives based on the ordering of a and b .

Proposition (Canonical Criterion). *For the triple $(a, b, c) = (m^2 - n^2, n(2m - n), m^2 - mn + n^2)$:*

$$a > b \iff m > 2n, \quad a = b \iff m = 2n, \quad a < b \iff m < 2n.$$

Proof. $a > b \iff m^2 - n^2 > n(2m - n) \iff m^2 - n^2 > 2mn - n^2 \iff m^2 > 2mn \iff m > 2n.$ $\square$

Corollary (No Self-Conjugate Primitives). *The boundary $m = 2n$ gives $a = b$, but then $m + n = 3n \equiv 0 \pmod{3}$, so such triples are non-primitive. Thus **no primitive triple has $a = b$** —all primitive triples come in conjugate pairs.*

The conjugation symmetry has a clean parametric description.

Proposition (Conjugation Symmetry). *The parameter pairs (m, n) and $(m, m - n)$ produce triples (a, b, c) and (b, a, c) . This corresponds to $\gamma \leftrightarrow \bar{\gamma}$ in $\mathbb{Z}[\omega]$.*

Proof. For $(m, m - n)$: $a' = m^2 - (m - n)^2 = 2mn - n^2 = n(2m - n) = b$, and similarly $b' = a.$ $\square$

Definition (Canonical Primitive Triple). A primitive E-P triple is **canonical** (ctrim) if $a > b$. The canonical primitives are parametrised by:

$$\{(m, n) : \gcd(m, n) = 1, m > 2n > 0, m + n \not\equiv 0 \pmod{3}\}.$$

Each canonical primitive has a unique **conjugate** (b, a, c) arising from $(m, m - n)$.

8 The Projective Viewpoint and Fibration

The six-fold symmetry suggests passing to projective coordinates, where the unit group action becomes trivial. The classification then emerges from a fibration of the projective parameter space. In projective coordinates $[A : B : C]$, scaling is absorbed: $[ka : kb : kc] = [a : b : c]$. The curve $\mathcal{E}$ becomes:

$$\mathcal{E} : A^2 - AB + B^2 = C^2 \quad \subset \mathbb{P}^2.$$

A triple (a, b, c) is **primitive** iff it is the minimal integer representative of its projective equivalence class.

8.1 The Parameter Space as $\mathbb{P}^1(\mathbb{Q})$

The slope $t = n/m$ lives in $\mathbb{Q} \cup \{\infty\} \cong \mathbb{P}^1(\mathbb{Q})$. The parametrisation becomes:

$$\Phi : \mathbb{P}^1(\mathbb{Q}) \longrightarrow \mathcal{E}(\mathbb{Q}), \quad [m : n] \longmapsto [m^2 - n^2 : n(2m - n) : m^2 - mn + n^2].$$

This is a bijection onto $\mathcal{E}(\mathbb{Q}) \setminus \{[-1 : 0 : 1]\}$, since a smooth plane conic with a rational point is isomorphic to $\mathbb{P}^1$ via the pencil of lines through that point.[\[4, 7\]](#)

8.2 The Parity Fibration

The key geometric structure is a fibration of the parameter space over the three-element group.

Definition. The **parity map** is:

$$\pi : \mathbb{P}^1(\mathbb{Z}) \rightarrow \mathbb{Z}_3, \quad [m : n] \mapsto (m + n) \bmod 3.$$

This map fibres the parameter space over the three-element group $\mathbb{Z}_3 \cong \mathbb{Z}[\omega]/\langle\beta\rangle$.

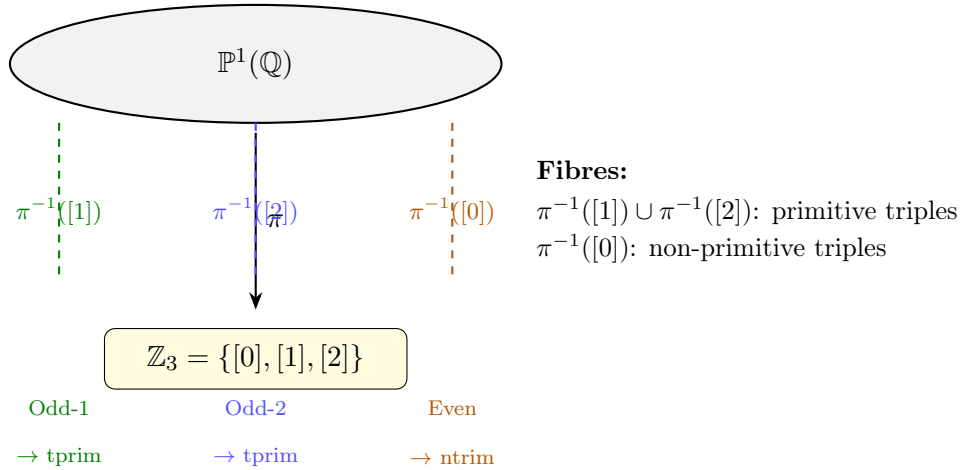


Figure 7: The reduced parameter set $\{(m, n) \in \mathbb{Z}^2 : \gcd(m, n) = 1\} / \{\pm 1\}$ admits a natural three-way partition via $(m + n) \bmod 3$, mirroring reduction of $\gamma = m + n\omega$ modulo $(\beta) \cong \mathbb{Z}/3\mathbb{Z}$.

The parity map has a simple linear algebra Matrix description.

Proposition. The map π is induced by the linear functional:

$$\ell : \mathbb{Z}^2 \longrightarrow \mathbb{Z}_3, \quad \ell(m, n) = m + n \bmod 3.$$

In matrix form: $\ell = \begin{pmatrix} 1 & 1 \end{pmatrix}$. The even fibre $\pi^{-1}([0])$ is the kernel:

$$\ker(\ell) = \{(m, n) \in \mathbb{Z}^2 : m + n \equiv 0 \pmod{3}\},$$

a sublattice of index 3 in $\mathbb{Z}^2$.

Equivalently, π is the reduction of the Eisenstein integer $\gamma = m + n\omega$ modulo the ideal (β) , since $\gamma \equiv m + n \pmod{\beta}$. [9, 1]

The fibration as a coset correspondence connects the parameter space to the Eisenstein parity classification.

Theorem (Coset Correspondence). *The following diagram commutes:*

$$\begin{array}{ccc} \mathbb{P}^1(\mathbb{Q}) & \xrightarrow{\Phi} & \mathcal{E}(\mathbb{Q}) \\ \pi \downarrow & & \\ \mathbb{Z}_3 & \xrightarrow{\sim} & \mathbb{Z}[\omega]/\langle\beta\rangle \end{array}$$

where Φ is the parametrisation and the bottom isomorphism sends $[k] \mapsto [k]_\beta$.

The parity of the generator $\gamma = m + n\omega$ in the sense of Hadi et al. equals the fibre of $[m : n]$ under π .

The fibration structure is visible directly in the parameter space. Figure 8 shows the mod-3 partition of coprime pairs (m, n) with $m > n > 0$, with the boundary $m = 2n$ separating canonical from conjugate primitives.

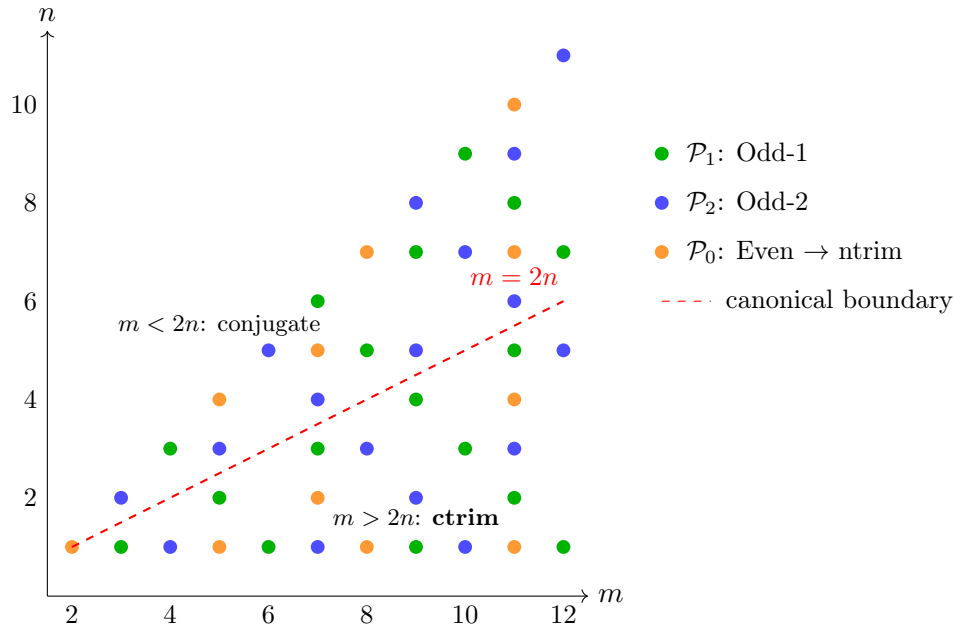


Figure 8: The mod-3 partition of the parameter space $\{(m, n) : \gcd(m, n) = 1, m > n > 0\}$. Orange points (even fibre) yield non-primitive triples. Green and blue points (odd fibres) yield primitives, with the line $m = 2n$ separating canonical ($a > b$) from conjugate ($a < b$).

The three parity classes are asymptotically equidistributed: as $M \rightarrow \infty$, each contains approximately $\frac{1}{3}$ of coprime pairs with $m \leq M$.

Geometric Meaning of Parity

The algebraic parity classification of Eisenstein integers is the fibration of the parameter space over $\mathbb{Z}_3$. The “even” coset $[0]_\beta$ corresponds to the kernel of the linear functional $\ell(m, n) = m + n$.

8.3 The Conjugation Involution

The conjugation $\gamma \mapsto \bar{\gamma}$ induces an involution on the parameter space.

Proposition. *Under $\gamma = m + n\omega \mapsto \bar{\gamma} = (m - n) - n\omega$, the parameter pair transforms as:*

$$(m, n) \mapsto (m, m - n).$$

In matrix form:

$$\begin{pmatrix} m \\ n \end{pmatrix} \mapsto \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} m \\ n \end{pmatrix} = \begin{pmatrix} m \\ m - n \end{pmatrix}.$$

This involution swaps $a \leftrightarrow b$ in the output triple.

Proof. We have $\bar{\omega} = -1 - \omega$, so:

$$\overline{m + n\omega} = m + n\bar{\omega} = m + n(-1 - \omega) = (m - n) - n\omega.$$

Writing this as $m' + n'\omega$ with $m' = m - n$ and $n' = -n$, we reparametrise to get $(m, m - n)$ in the standard form with $m > n > 0$. $\square$

Corollary. *The conjugation involution swaps the odd-1 and odd-2 fibres:*

$$\pi^{-1}([1]) \xleftrightarrow{\text{conj.}} \pi^{-1}([2]).$$

The even fibre $\pi^{-1}([0])$ is preserved (setwise, not pointwise).

This explains why both odd fibres yield primitive triples: they are exchanged by conjugation, which preserves primitivity.

8.4 The Canonical/Conjugate Refinement

Within the primitive class, we distinguish canonical representatives.

Definition. A primitive E-P triple is **canonical** (ctrim) if $a > b$. The parametrisation:

$$\mathcal{C} = \{(m, n) \in \mathcal{F} : m + n \not\equiv 0 \pmod{3}, m > 2n\}$$

produces exactly the canonical primitives.

Proposition. *The condition $m > 2n$ is equivalent to $a > b$. The boundary $m = 2n$ forces $m + n = 3n \equiv 0 \pmod{3}$, so lies in the even fibre.*

This gives the complete picture:

Region	Fibre	Constraint	Output
$m + n \equiv 0$	$\pi^{-1}([0])$	any	ntrim
$m + n \not\equiv 0, m > 2n$	$\pi^{-1}([1]) \cup \pi^{-1}([2])$	canonical	ctrim
$m + n \not\equiv 0, m < 2n$	$\pi^{-1}([1]) \cup \pi^{-1}([2])$	conjugate	tprim, $a < b$

9 The Complete Sieve

The parametrisation $(m, n) \mapsto (a, b, c)$ is governed by two independent structures:

1. A **mod-3 algebraic partition** coming from reduction of $\gamma = m + n\omega$ modulo (β) , separating even and odd generators.
2. A **geometric symmetry** of the hexagonal lattice, where conjugation exchanges a and b , allowing restriction to $m > 2n$ for canonical representatives.

These combine to produce a three-way filtration of all coprime parameter pairs (m, n) into:

ntrim, ctrim, and conjugate classes,

each occupying asymptotically one-third of the parameter space and is shown in Figure 9.

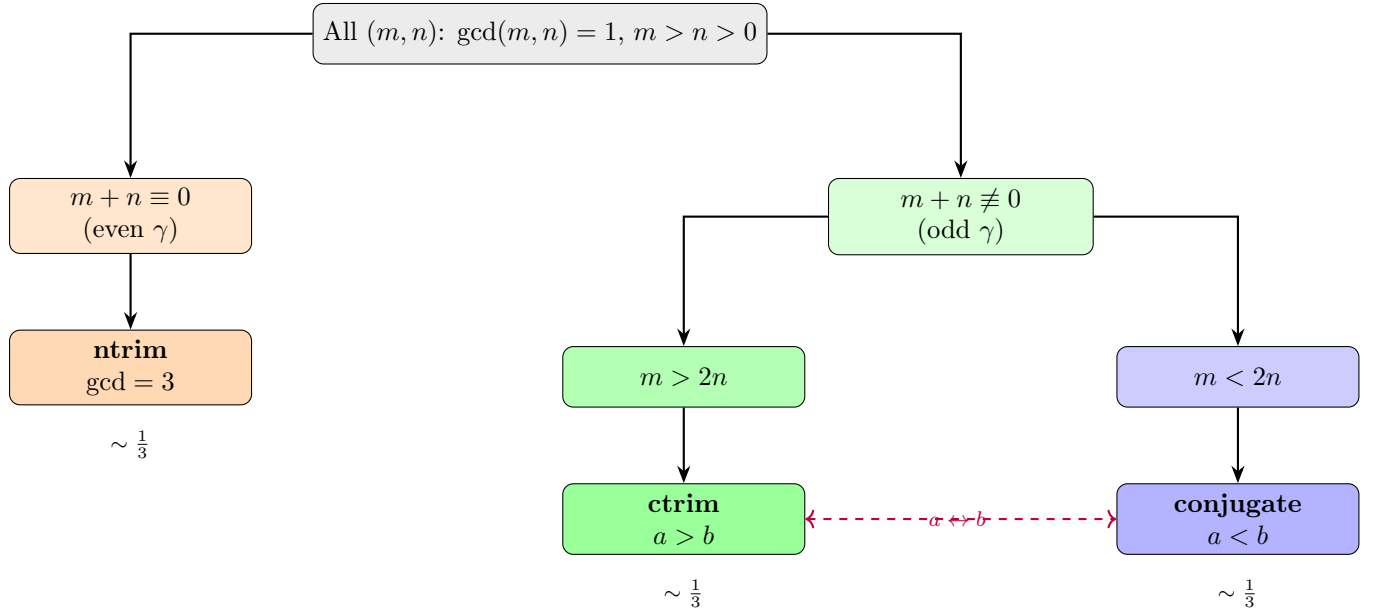


Figure 9: The complete sieve from parameters to the three classes.

The parity multiplication rules (Theorem 6.3) predict that odd generators of either type square to Odd-1, while even generators remain even as verified in Table 5.

Table 5: Parity inheritance under squaring: both Odd-1 and Odd-2 generators produce Odd-1 outputs, confirming Corollary 6.3

m	n	$m + n \bmod 3$	Gen. parity	$a + b \bmod 3$	Output parity
3	1	1	Odd-1	$13 \equiv 1$	Odd-1
3	2	2	Odd-2	$13 \equiv 1$	Odd-1
4	1	2	Odd-2	$22 \equiv 1$	Odd-1
4	3	1	Odd-1	$22 \equiv 1$	Odd-1
5	1	0	Even	$33 \equiv 0$	Even
5	2	1	Odd-1	$37 \equiv 1$	Odd-1
5	3	2	Odd-2	$37 \equiv 1$	Odd-1
5	4	0	Even	$33 \equiv 0$	Even

9.1 Complete Classification Tables

Table 6: Eisenstein–Pythagorean triples organised by lineage. Triples arise from generators $\gamma = m + n\omega$ with $\gcd(m, n) = 1$. The congruence $(m + n) \bmod 3$ determines *generator parity*: $m + n \equiv 0$ produces non-primitive triples (ntrim) with $\gcd(a, b, c) = 3$, while $m + n \not\equiv 0$ produces triple-primitive ones (tprim) with $\gcd(a, b, c) = 1$. Among primitives, the inequality $m \gtrless 2n$ distinguishes canonical triples ($a > b$) from their conjugates ($a < b$). Non-primitive triples also occur in conjugate pairs (visible as matching c -values), but we reserve the canonical/conjugate terminology for the primitive class.

m	n	a	b	c	$m + n$	$m : 2n$	Lineage
2	1	3	3	3	$3 \equiv 0$	$=$	ntrim
5	1	24	9	21	$6 \equiv 0$	$>$	ntrim
5	4	9	24	21	$9 \equiv 0$	$<$	ntrim
7	2	45	24	39	$9 \equiv 0$	$>$	ntrim
7	5	24	45	39	$12 \equiv 0$	$<$	ntrim
8	1	63	15	57	$9 \equiv 0$	$>$	ntrim
8	7	15	63	57	$15 \equiv 0$	$<$	ntrim
3	1	8	5	7	$4 \equiv 1$	$>$	ctrim
4	1	15	7	13	$5 \equiv 2$	$>$	ctrim
5	2	21	16	19	$7 \equiv 1$	$>$	ctrim
6	1	35	11	31	$7 \equiv 1$	$>$	ctrim
7	1	48	13	43	$8 \equiv 2$	$>$	ctrim
7	3	40	33	37	$10 \equiv 1$	$>$	ctrim
9	1	80	17	73	$10 \equiv 1$	$>$	ctrim
9	2	77	32	67	$11 \equiv 2$	$>$	ctrim
3	2	5	8	7	$5 \equiv 2$	$<$	conj. of (8, 5, 7)
4	3	7	15	13	$7 \equiv 1$	$<$	conj. of (15, 7, 13)
5	3	16	21	19	$8 \equiv 2$	$<$	conj. of (21, 16, 19)
6	5	11	35	31	$11 \equiv 2$	$<$	conj. of (35, 11, 31)
7	4	33	40	37	$11 \equiv 2$	$<$	conj. of (40, 33, 37)
7	6	13	48	43	$13 \equiv 1$	$<$	conj. of (48, 13, 43)
9	7	32	77	67	$16 \equiv 1$	$<$	conj. of (77, 32, 67)
9	8	17	80	73	$17 \equiv 2$	$<$	conj. of (80, 17, 73)

Remark (Conjugate structure in the non-primitive class). The conjugation symmetry $(m, n) \leftrightarrow (m, m - n)$, corresponding to $\gamma \leftrightarrow \bar{\gamma}$, operates on *all* generators regardless of parity. Thus non-primitive triples also occur in $a \leftrightarrow b$ pairs: (24, 9, 21) and (9, 24, 21) both have $c = 21$ and $\gcd(a, b, c) = 3$. However, the canonical/conjugate distinction is only meaningful for primitive triples, where we need a unique representative. For non-primitives, dividing through by 3 yields the *reduced triple*: $(24, 9, 21) \rightarrow (8, 3, 7)$, $(9, 24, 21) \rightarrow (3, 8, 7)$. These reduced triples are themselves a conjugate pair of primitives. Thus each non-primitive pair corresponds, after reduction, to a primitive conjugate pair. The boundary case $(m, n) = (2, 1)$ gives $(a, b, c) = (3, 3, 3)$ being the unique *self-conjugate* non-primitive triple, arising from $m = 2n$ which forces $a = b$.

10 Conclusion

We have established a complete classification of Eisenstein-Pythagorean triples via a three-way partition:

1. **Non-primitive (ntrim):** $m + n \equiv 0 \pmod{3}$ — the “even fibre” of the parameter space.
2. **Canonical primitive (ctrim):** $m + n \not\equiv 0 \pmod{3}$ and $m > 2n$ — the natural representatives with $a > b$.
3. **Conjugate primitive:** $m + n \not\equiv 0 \pmod{3}$ and $m < 2n$ — the $a \leftrightarrow b$ swap of canonicals.

The geometric insight is that the parameter space $\mathbb{P}^1(\mathbb{Q})$ **fibres over** the three-element group $\mathbb{Z}_3 \cong \mathbb{Z}[\omega]/\langle\beta\rangle$, with two fibres yielding primitive triples and one failing. The boundary case $m = 2n$ (where $a = b$) necessarily lies in the even fibre, so **no self-conjugate primitive triple exists**.

10.1 Connection to Quadratic Reciprocity

The splitting behaviour of primes in $\mathcal{O}_K$ determines which integers can appear as the “hypotenuse” c in primitive triples. We first illustrate with concrete examples.

Example (Prime Behaviour of Gaussian Ring). In $\mathbb{Z}[i]$, consider how small rational primes factor in the Gaussian integers:

p	Factorisation in $\mathbb{Z}[i]$	Behaviour	Reason
2	$(1+i)(1-i) = (1+i)^2 \cdot (-i)$	Ramifies	$2 \mid \text{disc}(-4)$
5	$(2+i)(2-i)$	Splits	$5 \equiv 1 \pmod{4}$
13	$(3+2i)(3-2i)$	Splits	$13 \equiv 1 \pmod{4}$
3	3 (remains prime)	Inert	$3 \equiv 3 \pmod{4}$
7	7 (remains prime)	Inert	$7 \equiv 3 \pmod{4}$

Note: $(1+i)$ and $(1-i)$ are associates (differing by unit $-i$), so 2 ramifies rather than splits.

Example (Prime Behaviour of Eisenstein Ring). The same primes in the Eisenstein integers:

p	Factorisation in $\mathbb{Z}[\omega]$	Behaviour	Reason
3	$(1-\omega)(1-\bar{\omega}) = (1-\omega)^2 \cdot (-\omega^2)$	Ramifies	$3 \mid \text{disc}(-3)$
7	$(3+\omega)(3+\bar{\omega}) = (3+\omega)(2-\omega)$	Splits	$7 \equiv 1 \pmod{3}$
13	$(4+\omega)(4+\bar{\omega}) = (4+\omega)(3-\omega)$	Splits	$13 \equiv 1 \pmod{3}$
2	2 (remains prime)	Inert	$2 \equiv 2 \pmod{3}$
5	5 (remains prime)	Inert	$5 \equiv 2 \pmod{3}$

Note: $(1-\omega)$ and $(1-\bar{\omega}) = (1-\omega^2) = (2+\omega)$ are associates, so 3 ramifies. (See Appendix.)

Example (Primes that Split in Both Rings). The prime 13 splits in both:

- In $\mathbb{Z}[i]$: $13 = (3+2i)(3-2i)$, and indeed $13 = 3^2 + 2^2$.
- In $\mathbb{Z}[\omega]$: $13 = (4+\omega)(3-\omega)$, and indeed $13 = 4^2 - 4 \cdot 1 + 1^2 = 3^2 - 3 \cdot (-1) + (-1)^2$.

This occurs because $13 \equiv 1 \pmod{4}$ and $13 \equiv 1 \pmod{3}$, i.e., $13 \equiv 1 \pmod{12}$.

The pattern is clear: a prime p is representable by the norm form precisely when it splits or ramifies. We now state this precisely.

Definition (Legendre Symbol). For an odd prime p and integer a with $p \nmid a$, the **Legendre symbol** is

$$\left(\frac{a}{p}\right) = \begin{cases} +1 & \text{if } a \equiv x^2 \pmod{p} \text{ for some } x, \\ -1 & \text{otherwise.} \end{cases}$$

Definition (Prime Behaviour in Quadratic Fields). Let $K = \mathbb{Q}(\sqrt{d})$ be a quadratic field with ring of integers $\mathcal{O}_K$. For a rational prime p , the principal ideal $(p) \subset \mathcal{O}_K$ factors in one of three ways:

1. **Splits:** $(p) = \mathfrak{p}\bar{\mathfrak{p}}$ with $\mathfrak{p} \neq \bar{\mathfrak{p}}$, where $\mathfrak{p}, \bar{\mathfrak{p}}$ are distinct prime ideals of norm p .
2. **Ramifies:** $(p) = \mathfrak{p}^2$, where $\mathfrak{p}$ is a prime ideal of norm p .
3. **Inert:** $(p) = \mathfrak{p}$ remains prime, with $N(\mathfrak{p}) = p^2$.

Here $\mathfrak{p}$ denotes a prime ideal of $\mathcal{O}_K$, and $\bar{\mathfrak{p}}$ its conjugate under $\sqrt{d} \mapsto -\sqrt{d}$.

Theorem (Splitting Criterion [5]). Let $K = \mathbb{Q}(\sqrt{d})$ with d squarefree. For an odd prime p :

- p **ramifies** iff $p \mid d$;
- p **splits** iff $p \nmid d$ and $\left(\frac{d}{p}\right) = 1$;
- p **is inert** iff $\left(\frac{d}{p}\right) = -1$.

The **supplements to quadratic reciprocity** determine the Legendre symbols for our cases:

Proposition ([5]). For an odd prime p :

$$\left(\frac{-1}{p}\right) = \begin{cases} +1 & \text{if } p \equiv 1 \pmod{4}, \\ -1 & \text{if } p \equiv 3 \pmod{4}; \end{cases} \quad \left(\frac{-3}{p}\right) = \begin{cases} +1 & \text{if } p \equiv 1 \pmod{3}, \\ -1 & \text{if } p \equiv 2 \pmod{3}. \end{cases}$$

Combining the examples, definitions, and criteria:

Corollary (Summary: Prime Behaviour in Both Rings).

	Gaussian $\mathbb{Z}[i]$	Eisenstein $\mathbb{Z}[\omega]$
<i>Field</i>	$\mathbb{Q}(\sqrt{-1})$	$\mathbb{Q}(\sqrt{-3})$
<i>Discriminant</i>	-4	-3
<i>Ramified prime</i>	$\mathfrak{p} = (1+i)$ over $p = 2$	$\mathfrak{p} = (1-\omega)$ over $p = 3$
<i>Split primes</i>	$p \equiv 1 \pmod{4}$	$p \equiv 1 \pmod{3}$
<i>Inert primes</i>	$p \equiv 3 \pmod{4}$	$p \equiv 2 \pmod{3}$
<i>Examples:</i>		
<i>Splits</i>	5, 13, 17, 29, 37, ...	7, 13, 19, 31, 37, ...
<i>Inert</i>	3, 7, 11, 19, 23, ...	2, 5, 11, 17, 23, ...

A prime p is representable by the norm form iff p splits or ramifies. And in either case $(p) = \mathfrak{p}\bar{\mathfrak{p}}$ or $(p) = \mathfrak{p}^2$ gives $p = N(\mathfrak{p})$. This yields the valid hypotenuse values:

Valid Pythagorean c	2, 5, 13, 17, 25, 29, 37, 41, ... $(p = 2 \text{ or } p \equiv 1 \pmod{4})$
Valid Eisenstein c	3, 7, 13, 19, 31, 37, 43, 49, ... $(p = 3 \text{ or } p \equiv 1 \pmod{3})$

Remark. Fermat's Two-Square Theorem ($p = a^2 + b^2$ iff $p = 2$ or $p \equiv 1 \pmod{4}$) and its Eisenstein analogue ($p = a^2 - ab + b^2$ iff $p = 3$ or $p \equiv 1 \pmod{3}$) are thus instances of a single principle: *a prime is representable by the norm form iff it splits or ramifies*. Multiplicativity of the norm extends this to composite values.

Remark (Connection to Eisenstein-Safe Primes). The split primes $p \equiv 1 \pmod{3}$ admit a further refinement. When $(p-1)/6$ is prime—equivalently, $p = 6q + 1$ for prime q —the prime is called *Eisenstein-safe*. Such primes have cryptographic significance: the residue field $\mathbb{F}_p$ admits a subgroup of prime order $(p-1)/6$ after factoring out the six-element torsion subgroup μ_6 . See [?] for details.

Notably, classical safe primes ($p = 2q + 1$) satisfy $p \equiv 2 \pmod{3}$ and are therefore *inert* so they cannot appear as hypotenuse values c in primitive E-P triples. The two notions of safety are thus arithmetically complementary.

10.2 The Unifying Principle

The Gaussian and Eisenstein theories are not parallel accidents, rather they are instances of two distinct arithmetic principles governing norm forms in imaginary quadratic fields. We will separate them cleanly here.

A. The Representability Principle

Let $K = \mathbb{Q}(\sqrt{d})$ be an imaginary quadratic field with ring of integers $\mathcal{O}_K$ and norm N .

Proposition (Representability Principle). *Let p be a rational prime with $p \nmid d$. Then*

$$p = N(\alpha) \quad \text{for some } \alpha \in \mathcal{O}_K$$

if and only if p splits in $\mathcal{O}_K$. Moreover, the unique ramified prime (dividing d) is also representable.

Proof sketch. If p splits, $(p) = \mathfrak{p}\bar{\mathfrak{p}}$ with $N(\mathfrak{p}) = p$.

In a class-number-one ring, $\mathfrak{p}$ is principal: $\mathfrak{p} = (\alpha)$, giving $N(\alpha) = p$.

If p ramifies, $(p) = \mathfrak{p}^2$ with $N(\mathfrak{p}) = p$, again yielding a generator of norm p .

If p is inert, (p) remains prime with $N((p)) = p^2$, so no element of norm p exists. □

Specialisation. **Gaussian case** ($d = -1$):

$$\begin{array}{ll} \text{Splits} & \iff p \equiv 1 \pmod{4}, \\ \text{Ramifies} & p = 2, \\ \text{Inert} & \iff p \equiv 3 \pmod{4}. \end{array}$$

Eisenstein case ($d = -3$):

$$\begin{array}{ll} \text{Splits} & \iff p \equiv 1 \pmod{3}, \\ \text{Ramifies} & p = 3, \\ \text{Inert} & \iff p \equiv 2 \pmod{3}. \end{array}$$

Thus Fermat's Two-Square Theorem and its Eisenstein analogue are instances of the same splitting criterion, derived from quadratic reciprocity.

Splitting governs which primes may appear as hypotenuse values c . But a different question arises: given a solution $N(\alpha) = c^2$, when is the resulting triple *primitive*?

B. The Primitivity Principle

This is controlled not by splitting, but by *ramification*.

Let $\mathfrak{p}$ be the unique prime of $\mathcal{O}_K$ lying above the ramified rational prime p . Since p ramifies,

$$(p) = \mathfrak{p}^2.$$

Write $\alpha = \gamma^2$ with γ coordinate-primitive.

Proposition (Primitivity Principle). *Let $\mathcal{O}_K$ have class number 1. The integer coordinates of $\alpha = \gamma^2$ are coprime if and only if $\mathfrak{p} \nmid \gamma$.*

Proof. If $\mathfrak{p} \mid \gamma$, then $\mathfrak{p}^2 \mid \gamma^2$. Since $\mathfrak{p}^2 = (p)$, this implies p divides every coordinate of γ^2 .

Conversely, if p divides all coordinates of γ^2 , then $(p) \mid (\gamma^2)$. Since $(p) = \mathfrak{p}^2$ and $\mathcal{O}_K$ is a UFD (class number one), it follows that $\mathfrak{p} \mid \gamma$. $\square$

Parity as Reduction Modulo the Ramified Prime

Reduction modulo $\mathfrak{p}$ gives

$$\mathcal{O}_K/\mathfrak{p} \cong \mathbb{Z}/p\mathbb{Z}.$$

The “parity” of a generator is simply its residue class in this quotient.

Gaussian case: $\mathbb{Z}[i]/(1+i) \cong \mathbb{Z}/2\mathbb{Z}$ — binary parity.

Eisenstein case: $\mathbb{Z}[\omega]/(1-\omega) \cong \mathbb{Z}/3\mathbb{Z}$ — ternary parity.

Primitivity is equivalent to the residue being nonzero: $[\gamma] \neq [0]$.

Structural Synthesis

Two Principles & One Framework

Representability is controlled by **splitting**.

Primitivity is controlled by **ramification**.

Quadratic reciprocity determines splitting behaviour: which primes p satisfy $p = N(\alpha)$.

Ramification determines the primitivity obstruction: when $p \mid \gcd(\text{coordinates of } \gamma^2)$.

The Gaussian mod-2 condition and the Eisenstein mod-3 condition are more than analogies being specialisations of the **ramified prime principle**:

$$\text{Primitive triple} \iff \mathfrak{p} \nmid \gamma \iff [\gamma] \neq 0 \text{ in } \mathcal{O}_K/\mathfrak{p}$$

10.3 Summary

The Gaussian and Eisenstein integers are distinguished among imaginary quadratic rings by two properties:

1. **Class number** 1: ensuring unique factorisation and clean parametrisation of solutions.
2. **Extra units:** $|\mathcal{O}_K^\times| > 2$, giving geometric symmetry (4-fold and 6-fold).

These are the *only* imaginary quadratic fields with both properties. The square and hexagonal lattices are the unique regular lattices admitting a Euclidean domain structure with nontrivial rotational symmetry.

Table 7: The two exceptional imaginary quadratic rings

	Gaussian $\mathbb{Z}[i]$	Eisenstein $\mathbb{Z}[\omega]$
Field	$\mathbb{Q}(\sqrt{-1})$	$\mathbb{Q}(\sqrt{-3})$
Discriminant	-4	-3
Class number	1	1
Units	$\{\pm 1, \pm i\}$	$\{\pm 1, \pm \omega, \pm \omega^2\}$
Lattice symmetry	4-fold (square)	6-fold (hexagonal)

The classification of Eisenstein-Pythagorean triples is governed by the same pair of arithmetic principles as the classical Pythagorean theory:

Two Principles & One Framework

- **Representability** (which primes can be c ?) is controlled by **splitting**, determined by quadratic reciprocity.
- **Primitivity** (when is the triple coprime?) is controlled by **ramification**: $\mathfrak{p} \nmid \gamma \iff$ primitive.

The parity of a generator γ is its residue class in $\mathcal{O}_K/\mathfrak{p} \cong \mathbb{Z}_p$. The Gaussian case ($p = 2$, binary parity) and the Eisenstein case ($p = 3$, ternary parity) are the two manifestations of these principles in rings with geometric symmetry.

The Eisenstein case, with its three parity classes and canonical/conjugate distinction, reveals structure invisible in the simpler Gaussian world. The six-fold symmetry of the hexagonal lattice, the fibration over $\mathbb{Z}_3$, and the absence of self-conjugate primitive triples are all merely a consequence of the fact that $3 > 2$!

The Unity of the First Two Roots of Unity

	Gaussian	Eisenstein
Lattice	Square	Hexagonal
Unit curve	$X^2 + Y^2 = 1$	$X^2 - XY + Y^2 = 1$
Ramified prime $\mathfrak{p}$	$(1 + i)$	$(1 - \omega)$
$p = N(\mathfrak{p})$	2	3
Parity quotient	$\mathcal{O}/\mathfrak{p} \cong \mathbb{Z}_2$	$\mathcal{O}/\mathfrak{p} \cong \mathbb{Z}_3$
Primitivity	$[\gamma] \neq 0$ in $\mathbb{Z}_2$	$[\gamma] \neq 0$ in $\mathbb{Z}_3$
Canonical condition	$p > q$	$m > 2n$
Self-conjugate primitives	None	None
Representability: splitting $\longleftrightarrow$ Primitivity: ramification		

The theory of Hadi et al. [1] on primitive, even, and odd Eisenstein integers provides the algebraic foundation. Our contribution is threefold: a geometric interpretation of the primitivity sieve as a fibration over $\mathbb{Z}_3$; the identification of the canonical/conjugate distinction within the primitive class; and the recognition that the Gaussian and Eisenstein theories instantiate the same two arithmetic principles: that of the Representability Principle governing valid hypotenuse values, and the Primitivity Principle governing coprimality of triples.

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Appendix: Ramification in the Gaussian and Eisenstein Integers

We present explicitly the ramification of the rational primes 2 in $\mathbb{Z}[i]$ and 3 in $\mathbb{Z}[\omega]$. In each case, ramification occurs because the two conjugate factors are associates, and hence generate the same prime ideal.

A. Ramification of 2 in $\mathbb{Z}[i]$

Recall that $i^2 = -1$ and $\bar{i} = -i$. We compute:

$$2 = (1 + i)(1 - i).$$

The norm of $1 + i$ is

$$N(1 + i) = (1 + i)(1 - i) = 2.$$

Now observe that $1 - i$ differs from $1 + i$ by multiplication by the unit $-i$:

$$-i(1 + i) = -i - i^2 = -i + 1 = 1 - i.$$

Thus

$$1 - i = -i(1 + i),$$

so $1 - i$ and $1 + i$ are associates. Since associates generate the same ideal,

$$(1 - i) = (1 + i).$$

Therefore,

$$(2) = (1 + i)(1 - i) = (1 + i)^2.$$

Hence 2 does not split into two distinct prime ideals; rather,

$$(2) = \mathfrak{p}^2, \quad \mathfrak{p} = (1 + i).$$

Thus 2 ramifies in $\mathbb{Z}[i]$.

B. Ramification of 3 in $\mathbb{Z}[\omega]$

Recall that

$$\omega^2 = -1 - \omega, \quad \bar{\omega} = \omega^2.$$

We compute:

$$3 = (1 - \omega)(1 - \bar{\omega}).$$

First simplify:

$$1 - \bar{\omega} = 1 - \omega^2 = 1 - (-1 - \omega) = 2 + \omega.$$

Thus

$$3 = (1 - \omega)(2 + \omega).$$

The norm of $1 - \omega$ is

$$N(1 - \omega) = (1 - \omega)(1 - \bar{\omega}) = 3.$$

Now we show that $2 + \omega$ is an associate of $1 - \omega$. Compute:

$$-\omega^2(1 - \omega) = -\omega^2 + \omega^3.$$

Since $\omega^3 = 1$ and $\omega^2 = -1 - \omega$, we obtain:

$$-\omega^2 = 1 + \omega,$$

so

$$-\omega^2(1 - \omega) = (1 + \omega) + 1 = 2 + \omega.$$

Thus

$$2 + \omega = -\omega^2(1 - \omega),$$

and since $-\omega^2$ is a unit, $2 + \omega$ and $1 - \omega$ are associates. Therefore,

$$(2 + \omega) = (1 - \omega),$$

and

$$(3) = (1 - \omega)(2 + \omega) = (1 - \omega)^2.$$

Hence 3 ramifies in $\mathbb{Z}[\omega]$ with

$$(3) = \mathfrak{p}^2, \quad \mathfrak{p} = (1 - \omega).$$

Structural Comparison

	$\mathbb{Z}[i]$	$\mathbb{Z}[\omega]$
Ramified prime	2	3
Generator	$1 + i$	$1 - \omega$
Conjugate	$1 - i$	$1 - \bar{\omega}$
Associate relation	$1 - i = -i(1 + i)$	$2 + \omega = -\omega^2(1 - \omega)$
Ideal factorisation	$(2) = (1 + i)^2$	$(3) = (1 - \omega)^2$

In both cases, the conjugate factors are associates. Because they generate the same prime ideal, the rational prime does not split into two distinct primes but instead collapses into the square of a single prime ideal. This collapse is precisely ramification.